

Bash Script Tasks

1. Create a bash script to check if a directory is available or not.

```
[root@ip-172-31-83-252 ~]# vi check-directory.bash
[root@ip-172-31-83-252 ~]# chmod 755 check-directory.bash
[root@ip-172-31-83-252 ~]# cat check-directory.bash
#!/bin/bash

#Purpose: cheaking for directory exists or not
#Date: 9/11/2026
#Modificaton: 9/11/2026
#Owner: parvez

read -p " Enter directory path: " dir
if [ -d "$dir" ]
then
    echo "Directory exists."
else
    echo "Directory does not exist."
fi
[root@ip-172-31-83-252 ~]# ./check-directory.bash
Enter directory path: /root/sample
Directory exists.
[root@ip-172-31-83-252 ~]#
```

2. Create a bash script to create multiple files.

```
[root@ip-172-31-83-252 ~]# vi create-files.bash
[root@ip-172-31-83-252 ~]# chmod 755 create-files.bash
[root@ip-172-31-83-252 ~]# cat create-files.bash
#!/bin/bash

#Purpose: Create multiple files
#Date: 11/09/2026
#Modification: 11/09/2026
#Owner: parvez

touch file1.txt file2.txt file3.txt file4.txt file5.txt

echo "Multiple files created successfully."
[root@ip-172-31-83-252 ~]# ./create-files.bash
Multiple files created successfully.
[root@ip-172-31-83-252 ~]# ls -l
total 72
```

```
-rw-r--r--. 1 root root 0 Sep 11 11:01 file1.txt
-rw-r--r--. 1 root root 0 Sep 11 11:01 file2.txt
-rw-r--r--. 1 root root 0 Sep 11 11:01 file3.txt
-rw-r--r--. 1 root root 0 Sep 11 11:01 file4.txt
-rw-r--r--. 1 root root 0 Sep 11 11:01 file5.txt
-rw-r--r--. 1 root root 0 Sep 8 12:59 filr.bash
```

3. Create a bash script totake a backup of a directory.

```
[root@ip-172-31-83-252 ~]# ls -ld /root/sample
drwxr-xr-x. 2 root root 6 Sep 11 09:46 /root/sample
[root@ip-172-31-83-252 ~]# vi backup-directory.bash
[root@ip-172-31-83-252 ~]# chmod 755 backup-directory.bash
[root@ip-172-31-83-252 ~]# cat backup-directory.bash
#!/bin/bash

#Purpose: Take backup of a directory
#Date: 11/09/2026
#Modification: 11/09/2026
#Owner: parvez

tar -czf /root/sample-backup.tar.gz /root/sample

echo "Directory backup created successfully."
[root@ip-172-31-83-252 ~]# ./backup-directory.bash
tar: Removing leading '/' from member names
Directory backup created successfully.
[root@ip-172-31-83-252 ~]# ls -lh /root/sample-backup.tar.gz
-rw-r--r--. 1 root root 115 Sep 11 11:59 /root/sample-backup.tar.gz
[root@ip-172-31-83-252 ~]# |
```

4. Create a bash script toinstall Nginx on an EC2 server.

```
[root@ip-172-31-83-252 ~]# vi install-nginx.bash
[root@ip-172-31-83-252 ~]# chmod 755 install-nginx.bash
[root@ip-172-31-83-252 ~]# cat install-nginx.bash
#!/bin/bash

#Purpose: Install Nginx on EC2 server
#Date: 11/09/2026
#Modification: 11/09/2026
#Owner: parvez

dnf install nginx -y

systemctl start nginx

systemctl enable nginx

echo "Nginx installed and started successfully."
[root@ip-172-31-83-252 ~]# ./install-nginx.bash
Last metadata expiration check: 2:47:21 ago on Fri Sep 11 09:31:39 2026.
Dependencies resolved.

```

Package	Architecture	Version	Repository	Size
Installing:				
nginx	x86_64	1:1.30.4-1.amzn2023.0.1	amazonlinux	34 k
Installing dependencies:				
gperf-tools-libs	x86_64	2.9.1-1.amzn2023.0.3	amazonlinux	308 k
libumwind	x86_64	1.4.0-5.amzn2023.0.3	amazonlinux	66 k
nginx-core	x86_64	1:1.30.4-1.amzn2023.0.1	amazonlinux	711 k
nginx-filesystem	noarch	1:1.30.4-1.amzn2023.0.1	amazonlinux	10 k
nginx-mime-types	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	21 k

```
Transaction Summary

```

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

5. Create a bash script to install Apache Tomcat on an EC2 server.

```
[root@ip-172-31-83-252 opt]# vi install-tomcat.bash
[root@ip-172-31-83-252 opt]# vi install-tomcat.bash
[root@ip-172-31-83-252 opt]# chmod 755 install-tomcat.bash
[root@ip-172-31-83-252 opt]# cat install-tomcat.bash
#!/bin/bash

#Purpose: Install Apache Tomcat on EC2 server
#Date: 11/09/2026
#Modification: 11/09/2026
#Owner: parvez

dnf install java-17-amazon-corretto -y

cd /opt

wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
tar -xzf apache-tomcat-9.0.121.tar.gz

echo "Apache Tomcat installed successfully."
[root@ip-172-31-83-252 opt]# ./install-tomcat.bash
Last metadata expiration check: 3:30:39 ago on Fri Sep 11 09:31:39 2026.
Package java-17-amazon-corretto-1:17.0.20+8-1.amzn2023.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
--2026-09-11 13:02:19-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13178703 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.121.tar.gz'

apache-tomcat-9.0.121.tar.gz      100%[=====] 12.57M  5.85MB/s  in 2.1s

2026-09-11 13:02:22 (5.85 MB/s) - 'apache-tomcat-9.0.121.tar.gz' saved [13178703/13178703]

Apache Tomcat installed successfully.
[root@ip-172-31-83-252 opt]# ls -ld /opt/apache-tomcat-9.0.121
drwxr-xr-x. 9 root root 16384 Sep 11 13:02 /opt/apache-tomcat-9.0.121
[root@ip-172-31-83-252 opt]# /opt/apache-tomcat-9.0.121/bin/startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Tomcat started.
[root@ip-172-31-83-252 opt]# ps -ef | grep tomcat
```

```

2026-09-11 13:02:22 (5.85 MB/s) - 'apache-tomcat-9.0.121.tar.gz' saved [13178703/13178703]

Apache Tomcat installed successfully.
[root@ip-172-31-83-252 opt]# ls -ld /opt/apache-tomcat-9.0.121
drwxr-xr-x. 9 root root 16384 Sep 11 13:02 /opt/apache-tomcat-9.0.121
[root@ip-172-31-83-252 opt]# /opt/apache-tomcat-9.0.121/bin/startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-83-252 opt]# ps -ef | grep tomcat
root      10238      1  8 13:07 pts/5    00:00:02 /usr/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-9.0.121/conf/logging.properties -Djava.util.logging.manager=org.apache.juli
ener.UMASK=0027 -Dignore.endorsed.dirs=-classpath /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar -Dcatalina.base=/opt/apache-tomcat-9.0.121 -Dcata
lina.home=/opt/apache-tomcat-9.0.121 -Djava.io.tmpdir=/opt/apache-tomcat-9.0.121/temp org.apache.catalina.startup.Bootstrap start
root      10269      7096  0 13:07 pts/5    00:00:00 grep --color=auto tomcat
[root@ip-172-31-83-252 opt]# AC
[root@ip-172-31-83-252 opt]# netstat -na | grep 8080
tcp6      0      0 :::8080          :::*              LISTEN
[root@ip-172-31-83-252 opt]# curl http://localhost:8080

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Apache Tomcat/9.0.121</title>
    <link href="favicon.ico" rel="icon" type="image/x-icon" />
    <link href="tomcat.css" rel="stylesheet" type="text/css" />
  </head>
  <body>
    <div id="wrapper">
      <div id="navigation" class="curved container">
        <span id="nav-home"><a href="https://tomcat.apache.org/">Home</a></span>
        <span id="nav-hosts"><a href="/docs/">Documentation</a></span>
        <span id="nav-config"><a href="/docs/config/">Configuration</a></span>
        <span id="nav-examples"><a href="/examples/">Examples</a></span>
        <span id="nav-wiki"><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Wiki</a></span>
        <span id="nav-lists"><a href="https://tomcat.apache.org/lists.html">Mailing Lists</a></span>
        <span id="nav-help"><a href="https://tomcat.apache.org/findhelp.html">Find Help</a></span>
        <br class="separator" />
      </div>
      <div id="asf-box">
        <h1>Apache Tomcat/9.0.121</h1>

```

 Not secure
 98.93.236.173:8080

[Home](#)
[Documentation](#)
[Configuration](#)
[Examples](#)
[Wiki](#)
[Mailing Lists](#)
[Find Help](#)

Apache Tomcat/9.0.121



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)

[Manager Application How-To](#)

[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)
[Realms & AAA](#)
[Examples](#)
[Servlet Specifications](#)

[First Web Application](#)
[JDBC DataSources](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users.

[Read more...](#)

[Release Notes](#)

Documentation

[Tomcat 9.0 Documentation](#)

[Tomcat 9.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

Getting Help

[FAQ](#) and [Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)
User support and discussion

[tomcat-dev](#)
Developer support and discussion for Apache Tomcat

6. Create a bash script to check if the Nginx service is running, if not running then script should start the service.

```
root@ip-172-31-83-252:~  
[root@ip-172-31-83-252 ~]# vi nginx-check.bash  
[root@ip-172-31-83-252 ~]# chmod 755 nginx-check.bash  
[root@ip-172-31-83-252 ~]# cat nginx-check.bash  
#!/bin/bash  
  
# Purpose: Check whether Nginx service is running or not  
# Date: 09/12/2026  
# Modification: 09/12/2026  
# Owner: parvez  
  
if systemctl is-active --quiet nginx  
then  
    echo "Nginx service is running."  
else  
    echo "Nginx service is not running. Starting Nginx..."  
    systemctl start nginx  
    echo "Nginx service started."  
fi  
[root@ip-172-31-83-252 ~]# ./nginx-check.bash  
Nginx service is running.  
[root@ip-172-31-83-252 ~]# ./nginx_check.bash  
-bash: ./nginx_check.bash: No such file or directory  
[root@ip-172-31-83-252 ~]# systemctl stop nginx  
[root@ip-172-31-83-252 ~]# systemctl is-active nginx  
inactive  
[root@ip-172-31-83-252 ~]# ./nginx-check.bash  
Nginx service is not running. Starting Nginx...  
Nginx service started.  
[root@ip-172-31-83-252 ~]# systemctl is-active nginx  
active  
[root@ip-172-31-83-252 ~]#
```

7. Create a bash script for a calculator.

```

[root@ip-172-31-83-252 ~]# vi calculator.bash
[root@ip-172-31-83-252 ~]# ls -l calculator.bash
-rw-r--r--. 1 root root 617 Sep 12 08:39 calculator.bash
[root@ip-172-31-83-252 ~]# chmod 755 calculator.bash
[root@ip-172-31-83-252 ~]# cat calculator.bash
#!/bin/bash

# Purpose: Create a simple calculator
# Date: 09/12/2026
# Modification: 09/12/2026
# Owner: parvez

read -p "Enter first number: " num1
read -p "Enter second number: " num2

echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"

read -p "Enter your choice: " choice

case $choice in
    1)
        echo "Result = $((num1 + num2))"
        ;;
    2)
        echo "Result = $((num1 - num2))"
        ;;
    3)
        echo "Result = $((num1 * num2))"
        ;;
    4)
        echo "Result = $((num1 / num2))"
        ;;
    *)
        echo "Invalid choice"
        ;;
esac
[root@ip-172-31-83-252 ~]# ./calculator.bash
Enter first number: 20
Enter second number: 5
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 1
Result = 25
[root@ip-172-31-83-252 ~]# |

```

8. Create a bash script to check if a directory exists, if not then create a directory.

```

root@ip-172-31-83-252:~
[root@ip-172-31-83-252 ~]# vi directory-check.bash
[root@ip-172-31-83-252 ~]# chmod 755 directory-check.bash
[root@ip-172-31-83-252 ~]# cat directory-check.bash
#!/bin/bash

# Purpose: Check whether a directory exists or not
# Date: 09/12/2026
# Modification: 09/12/2026
# Owner: parvez

read -p "Enter directory path: " dir

if [ -d "$dir" ]
then
    echo "Directory exists."
else
    echo "Directory does not exist. Creating directory..."
    mkdir -p "$dir"
    echo "Directory created successfully."
fi
[root@ip-172-31-83-252 ~]# ./directory-check.bash
Enter directory path: /tt/parvez
Directory does not exist. Creating directory...
Directory created successfully.
[root@ip-172-31-83-252 ~]# ./directory-check.bash
Enter directory path: /tt/parvez
Directory exists.
[root@ip-172-31-83-252 ~]# ll
total 96
-rwxr-xr-x. 1 root root 206 Sep 12 07:25 backup-directory.bash
-rwxr-xr-x. 1 root root 617 Sep 12 08:39 calculator.bash
-rwxr-xr-x. 1 root root 341 Sep  9 06:58 case.bash
-rwxr-xr-x. 1 root root 272 Sep  7 11:29 cheak.bash
-rwxr-xr-x. 1 root root 245 Sep 11 10:43 check-directory.bash
-rwxr-xr-x. 1 root root 178 Sep  7 10:23 count.bash
-rwxr-xr-x. 1 root root 206 Sep 11 11:00 create-files.bash
-rw-r--r--. 1 root root 314 Sep  7 09:30 curdate.bash
-rwxr-xr-x. 1 root root 362 Sep 12 08:48 directory-check.bash
-rw-r--r--. 1 root root  11 Sep  7 12:06 error.txt
-rwxr-xr-x. 1 root root 204 Sep  7 10:44 errorfile.bash
-rwxr-xr-x. 1 root root 103 Sep  9 06:26 file.bash
-rw-r--r--. 1 root root   0 Sep 11 11:01 file1.txt

```

9. Create a bash script to delete the last 3 lines of a file.

```

root@ip-172-31-83-252 ~
[root@ip-172-31-83-252 ~]# vi delete-last3.bash
[root@ip-172-31-83-252 ~]# cat delete-last3.bash
#!/bin/bash

# Purpose: Delete the last 3 lines of a file
# Date: 09/12/2026
# Modification: 09/12/2026
# Owner: parvez

read -p "Enter file path: " file

if [ -f "$file" ]
then
    sed -i '$d' "$file"
    sed -i '$d' "$file"
    sed -i '$d' "$file"
    echo "Last 3 lines deleted successfully."
else
    echo "File does not exist."
fi
[root@ip-172-31-83-252 ~]# ./delete-last3.bash
Enter file path: vi test.txt
File does not exist.

[root@ip-172-31-83-252 ~]# cat test.txt
line 1
line 2
line 3
line 4
line 5
[root@ip-172-31-83-252 ~]# ./delete-last3.bash
Enter file path: test.txt
Last 3 lines deleted successfully.
[root@ip-172-31-83-252 ~]# cat test.txt
line 1
line 2
[root@ip-172-31-83-252 ~]# |

```

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

```
[root@ip-172-31-83-252 ~]# chmod 755 cpu-monitor.bash
[root@ip-172-31-83-252 ~]# ./cpu-monitor.bash
Current disk usage: 27%
Disk usage is normal: 27%
[root@ip-172-31-83-252 ~]# ls -l cpu-monitor.bash
-rwxr-xr-x. 1 root root 998 Sep 12 09:41 cpu-monitor.bash
[root@ip-172-31-83-252 ~]# chmod 755 cpu-monitor.bash
[root@ip-172-31-83-252 ~]# ./cpu-monitor.bash
Current disk usage: 27%
Disk usage is normal: 27%
[root@ip-172-31-83-252 ~]#
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

```

root@ip-172-31-83-252:~
[root@ip-172-31-83-252 ~]# chmod 755 disk-monitor.bash
[root@ip-172-31-83-252 ~]# cat disk-monitor.bash
#!/bin/bash

# Purpose: Checking disk usage and sending email if usage is more than 80%
# Date: 12/09/2026
# Modification: 12/09/2026
# Owner: parvez

THRESHOLD=80
EMAIL="mohdparvez1407@gmail.com"

HOSTNAME=$(hostname)

DISK_USAGE=$(df -P / | awk 'NR==2 {gsub("%","", $5); print $5}')

echo "Current disk usage: ${DISK_USAGE}%"

if [ "$DISK_USAGE" -ge "$THRESHOLD" ]
then
    SUBJECT="HIGH DISK USAGE ALERT - $HOSTNAME"

    MESSAGE="WARNING!

Server: $HOSTNAME
Disk Usage: ${DISK_USAGE}%

Disk usage has reached or exceeded the ${THRESHOLD}% threshold.
Please check the server."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    if [ $? -eq 0 ]; then
        echo "Email notification sent successfully."
    else
        echo "Failed to send email notification."
        exit 1
    fi
else
    echo "Disk usage is normal: ${DISK_USAGE}%"
fi

[root@ip-172-31-83-252 ~]# ./disk-monitor.bash
Current disk usage: 27%
Disk usage is normal: 27%
[root@ip-172-31-83-252 ~]# vi disk-monitor.bash
[root@ip-172-31-83-252 ~]# ls -l disk-monitor.bash
-rwxr-xr-x. 1 root root 901 Sep 12 10:13 disk-monitor.bash

```

12. Bash script to monitor memory and if it is more than 80% then send email notification.

```

root@ip-172-31-83-252:~
[root@ip-172-31-83-252 ~]# vi memory-monitor.bash
[root@ip-172-31-83-252 ~]# chmod 755 memory-monitor.bash
[root@ip-172-31-83-252 ~]# ./memory-monitor.bash
Current memory usage: 19%
Memory usage is normal: 19%
[root@ip-172-31-83-252 ~]# ls -l memory-monitor.bash
-rwxr-xr-x. 1 root root 921 Sep 12 10:17 memory-monitor.bash
[root@ip-172-31-83-252 ~]#

```

Exercise: Crontab Entries

1. April 5th Midnight

```

root@ip-172-31-83-252:~
[root@ip-172-31-83-252 ~]# crontab -e
crontab: installing new crontab
[root@ip-172-31-83-252 ~]# crontab -l
0 0 5 4 * /root/test
[root@ip-172-31-83-252 ~]# |

```

2. 5th of Every November, January, June if it is a Thursday

```

root@ip-172-31-83-252:~
[root@ip-172-31-83-252 ~]# crontab -e
crontab: installing new crontab
[root@ip-172-31-83-252 ~]# crontab -l
0 0 5 4 * /root/test
0 0 5 1,6,11 * /root/thursday.bash
[root@ip-172-31-83-252 ~]# |

```

3. At 05 and 27th minutes of 9, 10, 11 hours every day

4. 34 minutes of 9th hour on 15th August

5. Every midnight

6. Every Weekend (Saturday night 11:59 PM)

7. After every reboot

```
root@ip-172-31-83-252:~  
[root@ip-172-31-83-252 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-83-252 ~]# crontab -l  
0 0 5 4 * /root/test  
0 0 5 1,6,11 * /root/thursday.bash  
5,27 9-11 * * * /root/test  
34 9 15 8 * /root/test  
0 0 * * * /root/test  
59 23 * * 6 /root/test  
59 23 * * 6 /root/test  
@reboot /root/test  
[root@ip-172-31-83-252 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-83-252 ~]# crontab -l  
0 0 5 4 * /root/test  
0 0 5 1,6,11 * /root/thursday.bash  
5,27 9-11 * * * /root/test  
34 9 15 8 * /root/test  
0 0 * * * /root/test  
59 23 * * 6 /root/test  
@reboot /root/test  
[root@ip-172-31-83-252 ~]#
```

```
[root@ip-172-31-83-252 ~]#  
[root@ip-172-31-83-252 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-83-252 ~]# crontab -l  
# 1. April 5th Midnight  
0 0 5 4 * /root/test  
  
# 2. 5th January, June, November if Thursday  
0 0 5 1,6,11 * /root/thursday.bash  
  
# 3. 05 and 27 minutes of 9, 10, 11 hours every day  
5,27 9-11 * * * /root/test  
  
# 4. 34 minutes of 9th hour on 15th August  
34 9 15 8 * /root/test  
  
# 5. Every Midnight  
0 0 * * * /root/test  
  
# 6. Every Saturday at 11:59 PM  
59 23 * * 6 /root/test  
  
# 7. After every reboot  
@reboot /root/test  
[root@ip-172-31-83-252 ~]#
```

